Nathan Ramirez

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Biomedical engineering and robotics senior (May 2027) who builds working devices: wearable sensors, robots and lab hardware. 1st place, BU BMES Design-A-Thon. Certified phlebotomist with patient-care experience; owner of a small business. Fluent in English and Spanish.

EDUCATION

Boston University, College of Engineering Boston, MA
B.S. Biomedical Engineering, Concentration in Robotics Expected May 2027
Relevant coursework: Device Design & Diagnostics, Introduction to Solid Biomechanics, Biomedical Signals and Control, Biomedical Measurements & Analysis, Intro to Robotics

Credentials: Certified Phlebotomy Technician (Northeast Medical Institute, 2025); MRI didactic coursework (ASRT, Jul-Aug 2026)

ENGINEERING PROJECTS

NeuroTrack Wearable Head Impact Monitor Spring 2026
- Built a wearable prototype on an ESP32 and BNO055 IMU; wrote C++ BLE firmware streaming quaternion orientation and linear acceleration at 100 Hz
- Wrote the Python receiver: 3D head rendering, impact detection at a 1.5 g threshold and CSV session logging; won 1st place at the BU BMES Design-A-Thon (sponsored by BTEC)

Automated Tissue Cutting Matrix, BU Senior Design Sep 2026 - Apr 2027
- Lead machining, prototyping and organization on a 5-person BME team with Dr. Tuan Tay's lab, building a device for repeatable multi-angle histological sectioning of mouse brains

Multi-Robot Autonomous Navigation Fall 2025
- Built a collision avoidance system for a 12-robot challenge in a 9 m x 5 m motion capture arena using A* path planning, Dubins curves and a 1.5-second prediction horizon
- Traced failures to root cause and fixed them: a coordinate-frame mismatch with the motion capture system, gridlock from overlapping detection radii, and paths blocked by finished robots

Dual-Arm Robotic Manipulator Summer 2025
- Designed modular grippers and joints in Onshape, 3D printed them, and upgraded control from an Arduino Uno to an ESP32 network over ESP-NOW
- Programmed joystick and servo control for independent arm motion; tested load capacity and positional feedback

EXPERIENCE

Gems Home Improvement, Contractor and Translator May 2022 - Feb 2025 | Hartford, CT
- Led onsite teams of 2-10 in lumber, welding and construction work, contributing to a major project finished one year ahead of schedule
- Designed and assembled load-bearing wall and roof frames from technical drawings and material specifications

Quest Diagnostics, Phlebotomist Jul 2025 - Sep 2025 | Manchester, CT
- Followed strict chain-of-custody procedures for labeling, pipetting, slide preparation, centrifugation and storage so each specimen was traceable and results stayed accurate
- Maintained a 95%+ first-stick success rate across 10-20+ patients daily by following collection protocols consistently

SKILLS

For this role: embedded firmware, sensor integration, rapid prototyping, root cause debugging, machining, patient-facing clinical work
CAD and fabrication: SolidWorks, Onshape, KiCad, 3D printing (Bambu Lab, OrcaSlicer), CNC-fabricated parts, soldering, welding
Embedded: ESP32, Arduino, BLE, ESP-NOW, IMU and ultrasonic sensors
Programming: C++, Python, MATLAB, Java, Git/GitHub
Lab and clinical: Phlebotomy, aseptic technique, pipetting, centrifugation, gel electrophoresis, blood slide preparation, vital signs
Languages: English (fluent), Spanish (fluent), Portuguese (proficient), Italian (working knowledge)